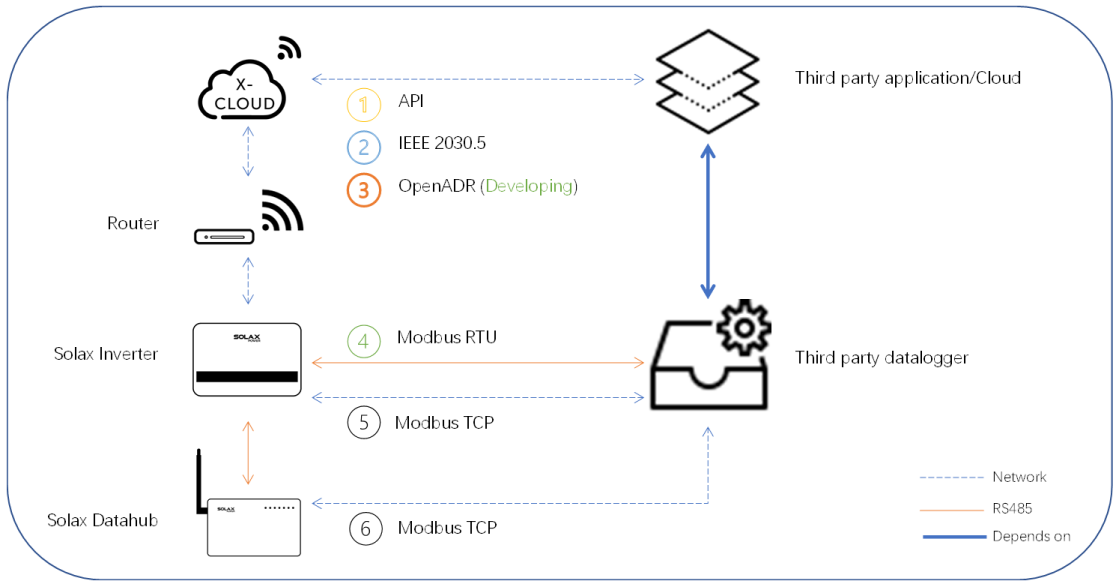


SolaX Power Eco-system

How to integrate to SolaX?



SolaX support multiple integration mechanisms for Eco-system customers' applications.



Topology

API

An Application Programming Interface (API) is a set of functions, procedures, methods or classes used by computer programs to request services from the operating system, software libraries or any other service providers running on the computer. A computer programmer uses the API to make application programs.

SolaX API

Monitoring API	Monitoring/Controlling API
<10 times/min	100 times/min
<10,000 times/day	Up to 100,000 times/day
Real time data (partially)	Real time data (fully)
×	History data
×	Alarm info
×	Comprehensive info
×	Firmware upgrade info
×	Remote control
×	Group operation
×	Inverter operation
×	Query

×	Device binding
For End-Users	For Installers and Agents
Free	Chargeable

Where to find SolaX API?

Monitoring/Controlling API for Installers and Distributors

Path: <https://www.solaxcloud.com> – Installer/Agent account login – Service – API

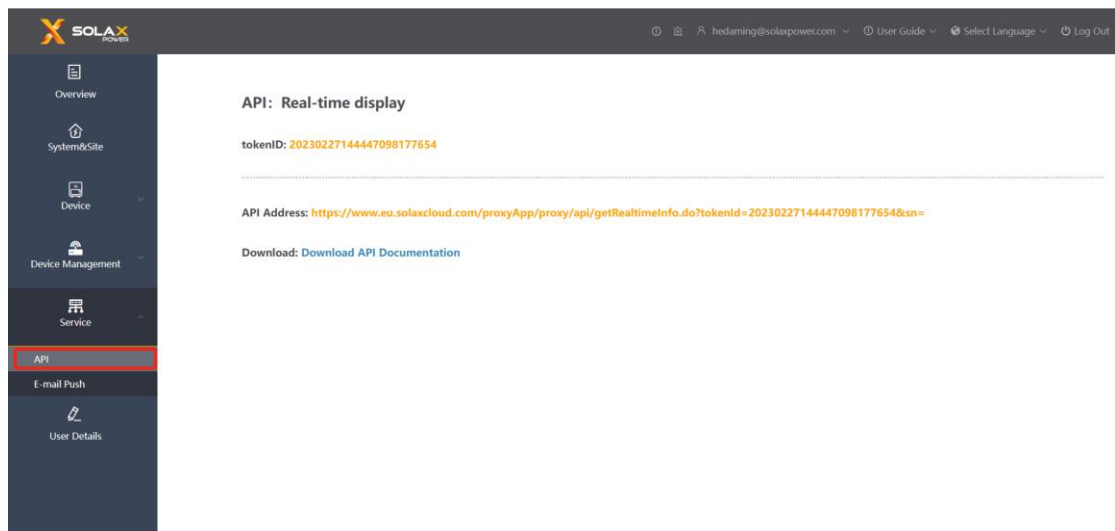
1. tokenID: It is a unique number for authentication be used in the API call command. Keep it safe.
2. Expire date: 3 months free trial. After the expiration, you can click the button to extend.
3. Plan Detail: Upgrade your plan according to the application.
4. API Address
5. Download: Download the Monitoring API directly. Contact your SolaX Representative for the Controlling API.

Note:

Document	Parameter	Refer to
Monitoring API	sn	Registration No. (communication module)
Controlling API	sn	Solax inverter serial number

Monitoring API for End-User

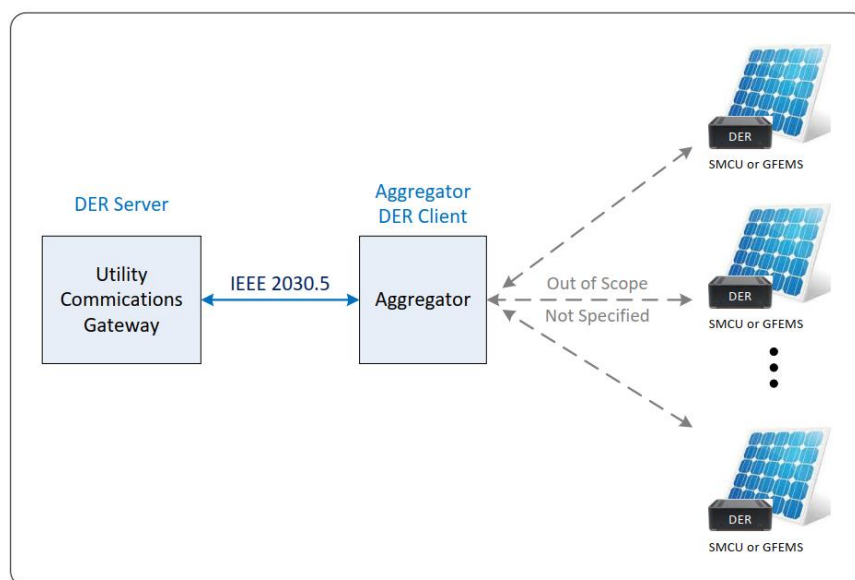
Path: <https://www.solaxcloud.com> – End-User account login – Service – API



IEEE 2030.5

Standard for Smart Energy Profile Protocol

This Standard defines an application profile which provides an interface between the smart grid and users. It enables management of the end user energy environment, including demand response, load control, price communication, distributed generation, energy storage, and electric vehicles as well as the support of additional commodities including water, natural gas, and steam.



OpenADR (developing)

The OpenADR Alliance was created to standardize, automate, and simplify Demand Response (DR) and Distributed Energy Resources (DER) to enable utilities and aggregators to cost-effectively manage growing energy demand & decentralized energy production, and customers to control their energy future. OpenADR is an open, highly secure, and two-way information exchange model and Smart Grid standard.

Modbus

Modbus is a data communications protocol originally published by Modicon (now Schneider Electric) in 1979 for use with its programmable logic controllers (PLCs). Modbus has become a de facto standard communication protocol and is now a commonly available means of connecting industrial electronic devices.

#	Model	Typ	Modbus TCP/IP		Modbus RTU
			Ethernet connection	Wifi connection	RS485
1	X1-Hybrid G3	Hybrid	Yes	No	No
2	X1-Retro Fit G3	battery	No	No	Yes
3	X1-Hybrid G4	Hybrid	No	Yes	Yes
4	X1-Retro Fit G4	Battery	No	Yes	Yes
5	X1 AC	battery	No	No	Yes
6	X3-Hybrid G2/3	Hybrid	Yes	No	No
7	X3-Retro Fit G2/3	Battery	No	No	Yes
8	X3 Hybrid G4	Hybrid	No	Yes	Yes
9	X3-Retro Fit G4	Battery	No	Yes	Yes